

Forensic Microscopy

Lab 4: Isotropy vs anisotropy

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Where can you find the slides?

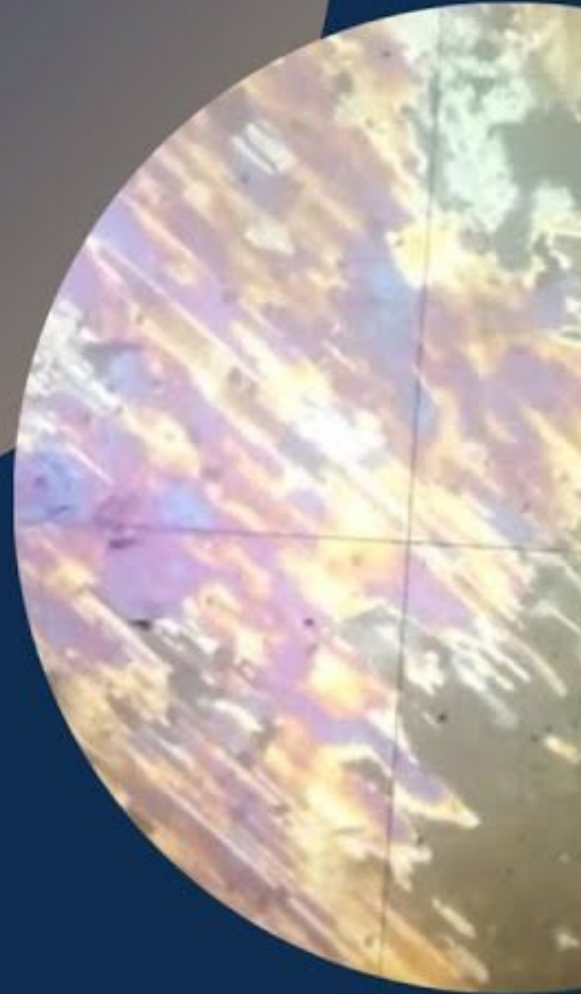


- <https://aritra-evolves.github.io/FIVS215.html>
- Canvas

(Grab a Microscope from the black closet, set up kohler)

Today's Agenda

- ❖ Polarization
- ❖ What is isotropy & anisotropy?
- ❖ Pleochroism
- ❖ Starches
- ❖ Calcite Rhombs
- ❖ Practice

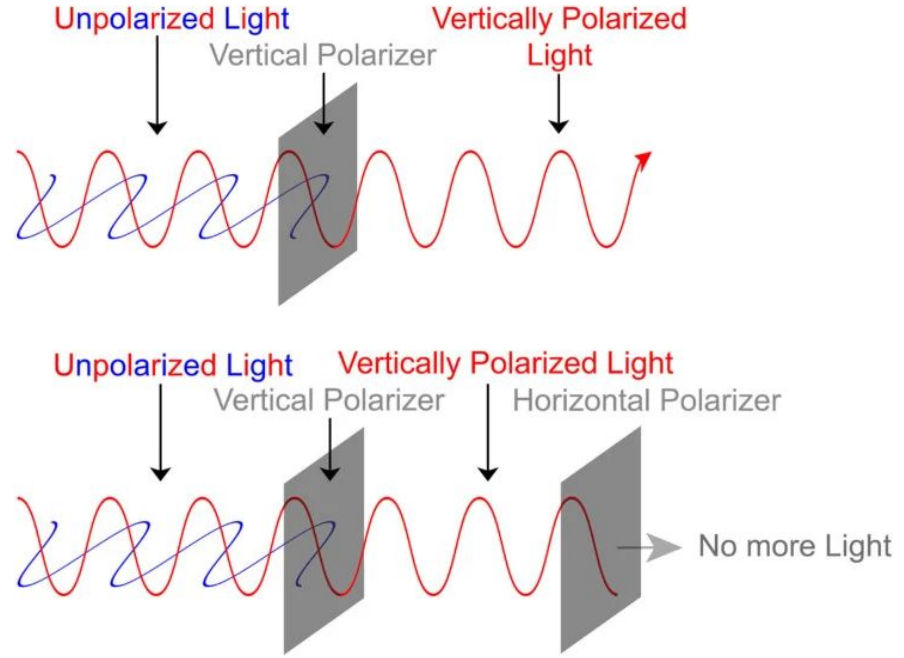


What is cross polarization?

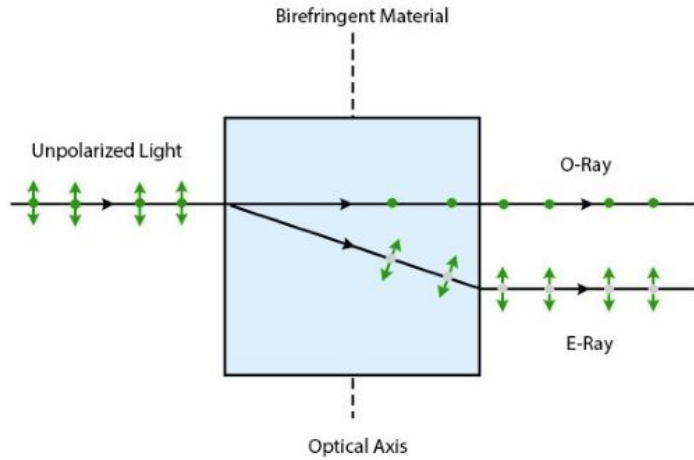
Polarization is the restricting of light to one vibrating plane.

Plane polarized light shows us one wave, cross polarized light uses a second polarizer to black out the light plane being shown.

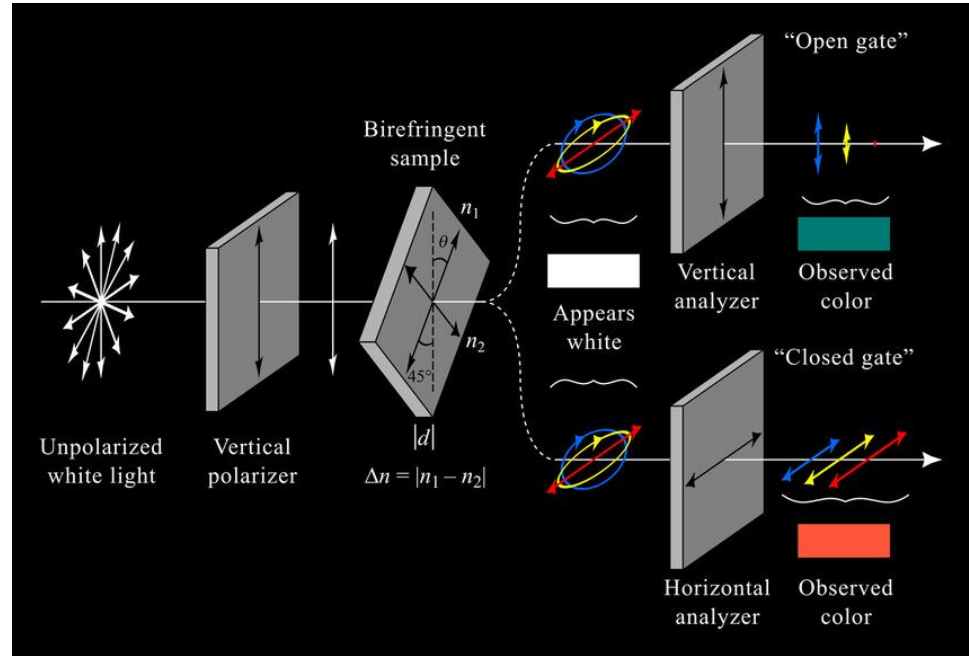
Allows us to view birefringence and retardation colors in samples and cancel out unwanted light in photographs.



Birefringence



<https://www.gophotonics.com/community/what-is-birefringence>

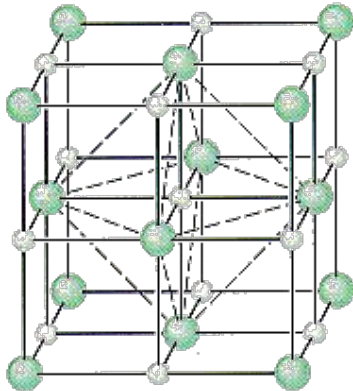


https://www.researchgate.net/figure/Color-online-A-schematic-representation-of-the-principle-of-polarization-filtered_fig1_362400155

Isotropy vs Anisotropy

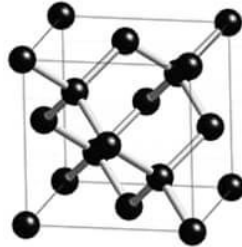
Isotropic

- No special effect on light
- Appears dark under cross polarized light
 - extinction
- Occurs in cubic crystalline structures
 - i.e. diamonds or NaCl



NaCl

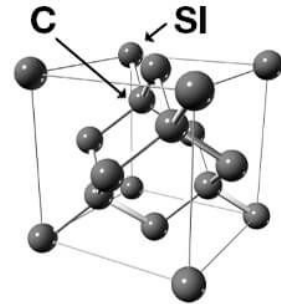
**DIAMONDS
ARE MADE
FROM CARBON
ATOMS [C]**



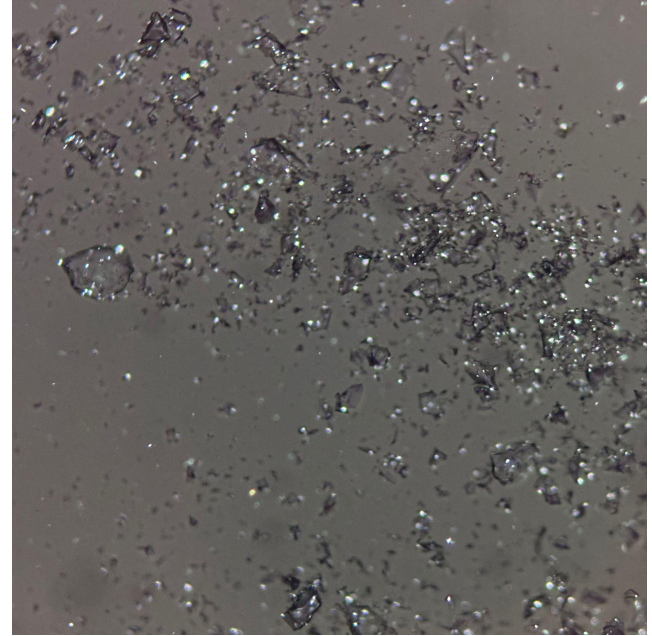
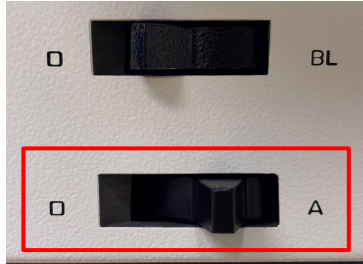
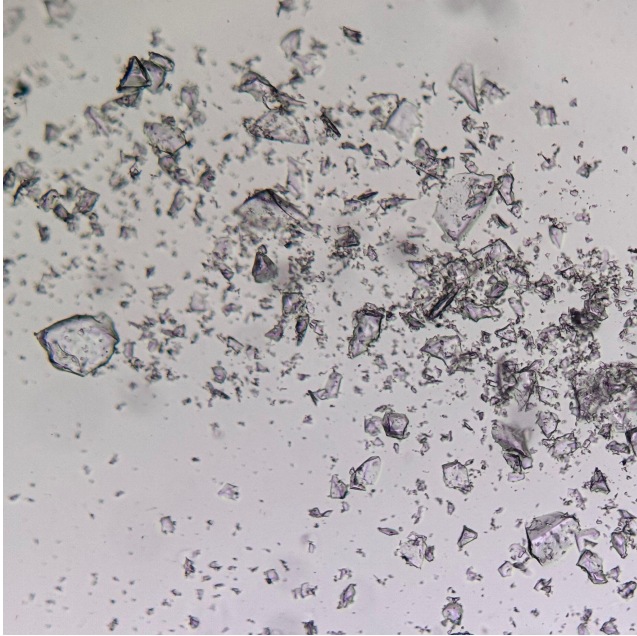
Anisotropic

- Effects light as it passes through
- Refracts light in different directions
- Appears colorful under cross polarized light
- Majority of substances
 - i.e. diamond substitutes, quartz, most precious gems

**MOISSANITES
ARE MADE
FROM SILICON
& CARBON
ATOMS [SiC]**

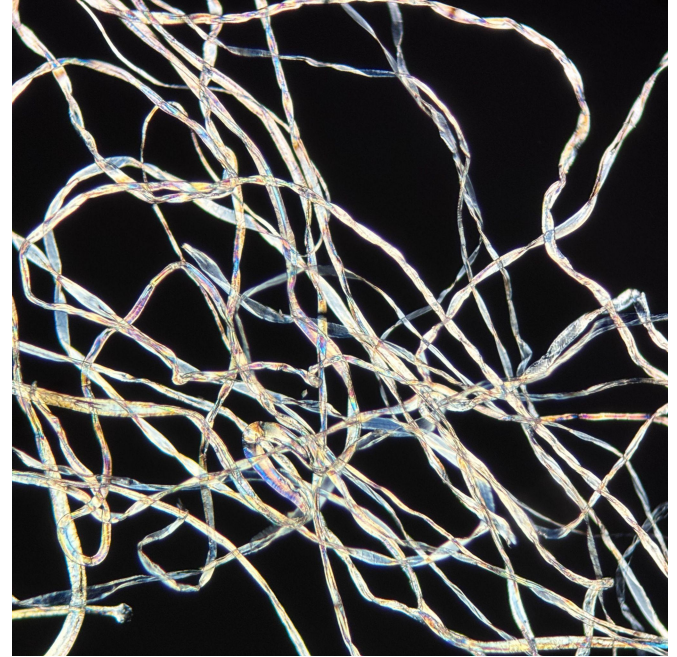
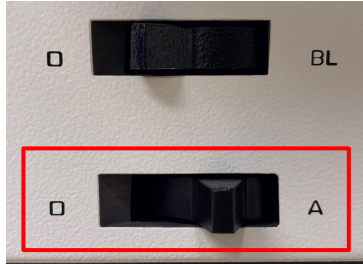
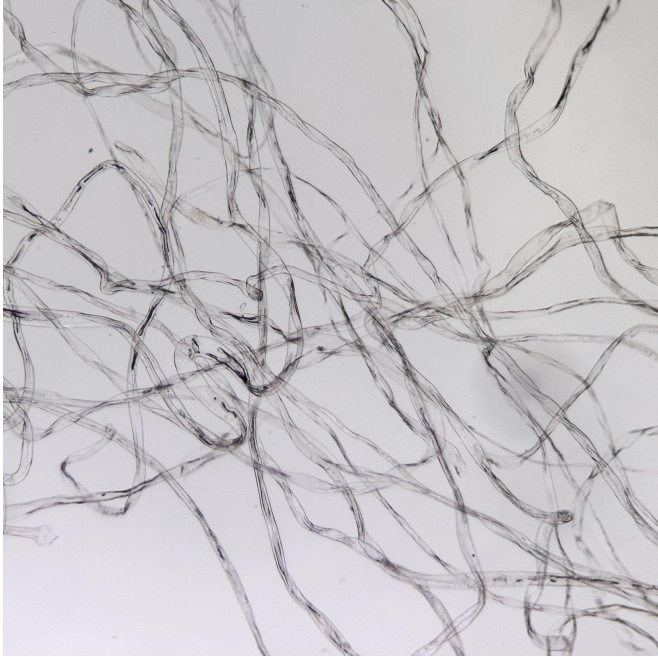


Isotropic material



Slide: locke glass (not to scale)

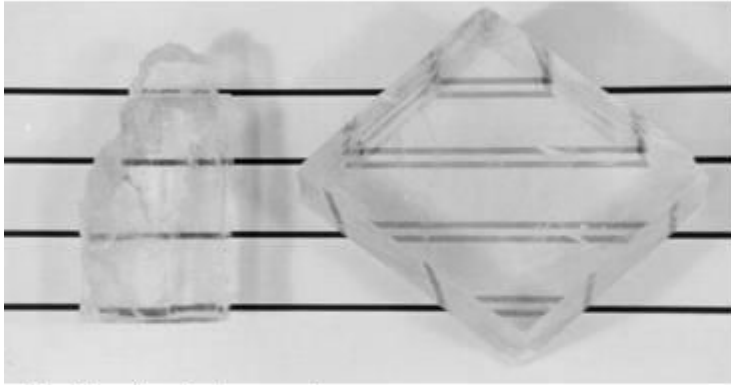
Anisotropic material



Slide: Cotton fiber (not to scale)

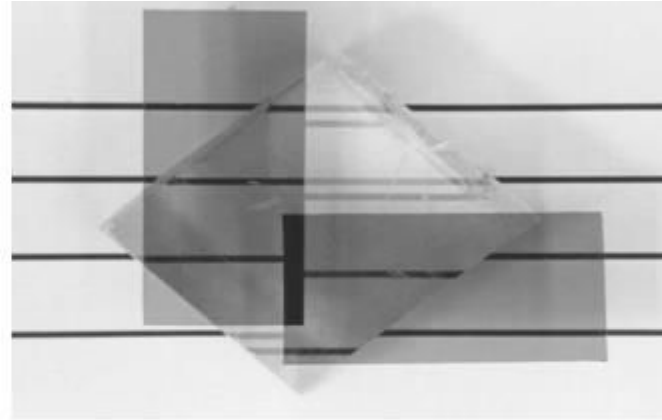
Calcite Rhombs

- Natural polarizer
- Splits rays in half - birefringence



Halite (cubic sodium chloride crystal, optically isotropic)

Calcite (optically anisotropic)

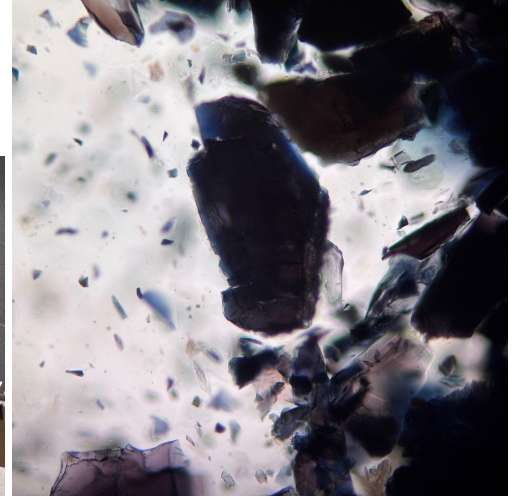
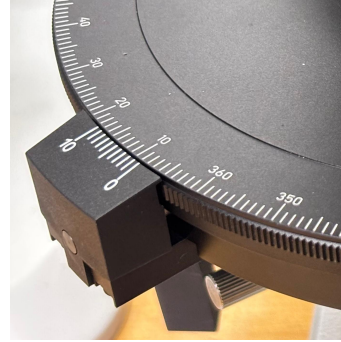
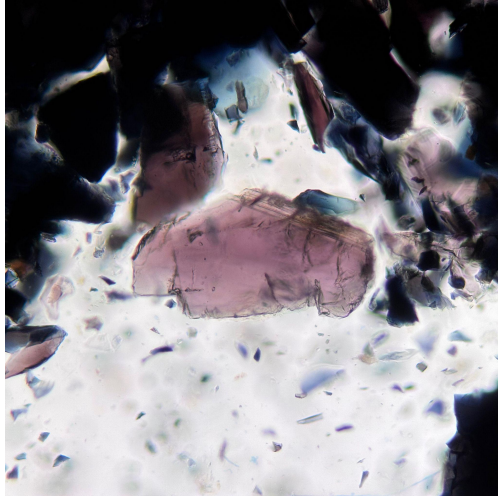
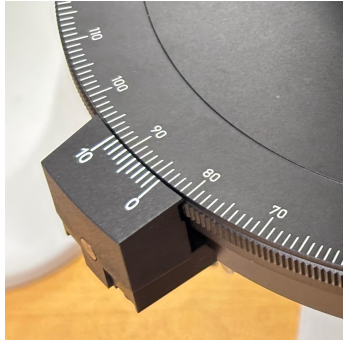


Calcite crystal with two polarizers at right angle to one another

Image:

https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.slideserve.com%2Fsonora%2Fbirefringence&psig=AOvVaw0KBVemSJk6Q7_SoNtZCjt7&ust=1758068741281000&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxqFwoTCLjpkMmC3l8DFQAAAAAAdAAAAABAV

Pleochroism

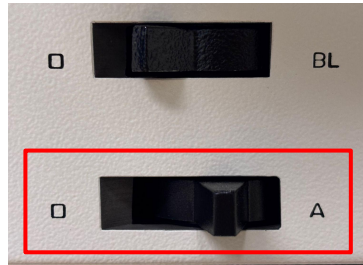
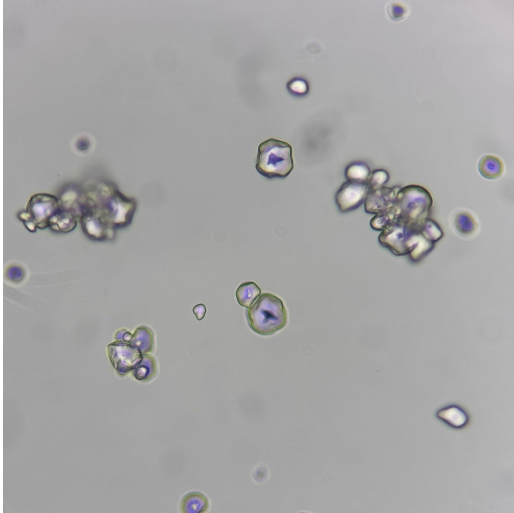


- As a sample rotates on the stage, plane polarized light passes through at different angles
- The sample will change color depending on the structure
- Isotropic: NEVER pleochroic
- Anisotropic: sometimes pleochroic
- Very common with minerals
 - i.e. asbestos, tourmaline, etc

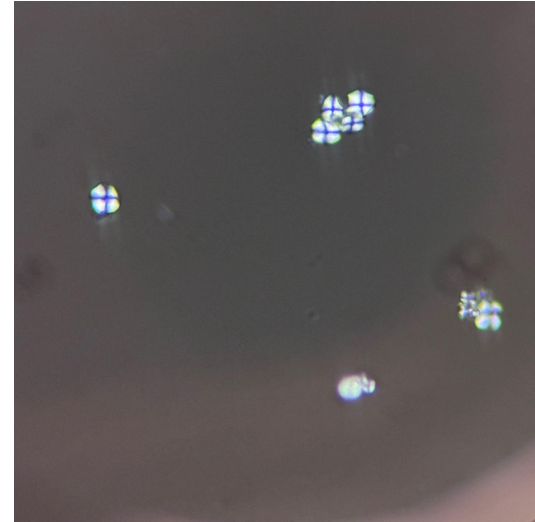
Slide: Tourmaline (not to scale)

Starches & Starch Grains

- Viewed under cross polars
- Creates a maltese cross
- Tells us if a starch is present
- You may need to increase magnification



Slide: Rice starch (not to scale)



Exercise: Determine isotropy and anisotropy

Slides from blue or white box, such as:

- Fibers
- NaCl
- Fiberglass
- Starch

Calcite rhomb

Handle the Tourmaline slide gently!

Return the slides to it's usual places

Reminders

- Review session for exam: This sunday (Sept. 20, 2026), 7.00 PM, over zoom, by Alli Findley.
(come prepared with doubts/questions)
- Link will be provided prior to session

See you next week!

